Neural Networks 201 (2026) 108885

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

TFMPHGNN: Two-Fold multi-perspective heterogeneous graph neural network for sentiment analysis

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ARTICLE INFO

Keywords:
Multi-perspective graphs
Sentiment analysis
Sentiment-emotion pair network
Heterogeneous graph neural network
Variational autoencoder.

ABSTRACT

Sentiment analysis remains challenging due to the complex, intertwined relationships among sentiment expressions, contextual cues, and emotional features distributed across heterogeneous data sources. Conventional deep learning and transformer-based models often treat sentiments as isolated units, failing to capture these rich, multi-perspective interactions. To address these limitations, this study introduces a Two-Fold Multi-Perspective Heterogeneous Graph Neural Network (TFMPHGNN) that jointly models sentiment, emotion, and contextual dependencies within a dual-stage heterogeneous graph framework. The first stage employs a meta-path-based encoder integrated with a capsule network to capture hierarchical semantic relationships among sentiment–emotion–context nodes, while the second stage utilizes a multi-channel graph convolutional network (MC-GCN) to learn complementary topological, semantic, and collaborative representations of sentiment–emotion pairs. A variational autoencoder (VAE) further denoises and refines latent embeddings. Experiments on the newly developed VaKSent-2025 corpus show that TFMPHGNN outperforms eight state-of-the-art graph-based baselines by 4.67% in accuracy, 2.7% in F1-micro, and 4.2% in F1-weighted, with statistical significance testing confirming the reliability of these gains. An extended ablation analysis further demonstrates that the collaborative fusion channel achieves 0.9387 accuracy, representing improvements of 7.5% and 7.9% over the topological-only and semantic-only channels, respectively, underscoring the synergistic value of integrating multiple graph perspectives. Collectively, these results indicate that TFMPHGNN effectively captures complex sentiment–emotion interdependencies and offers a robust, interpretable framework for fine-grained sentiment understanding.

1. Introduction

Sentiment analysis plays a crucial role in understanding public opinion and consumer behavior; however, traditional approaches exhibit significant limitations. Earlier sentiment-based methods often relied on labor-intensive, rule-based techniques that are time-consuming and prone to overfitting (Zhang et al., 2023). Conventional machine learning algorithms, including lexicon-based approaches combined with support vector machines and sentiment similarity measures, have been employed for supervised sentiment classification. Although these methods are generally straightforward and computationally efficient, they frequently struggle to capture the subtleties of human emotions and contextual dependencies. The rapid evolution of deep learning (DL) has led to substantial advances in sentiment analysis, enabling models to automatically extract high-quality features from sentiment-related data (Agbesi et al., 2025; Cheng et al., 2025). Many contemporary computational approaches leverage large-scale datasets that integrate textual, visual, and acoustic information. These methods employ heterogeneous network architectures to model the complex relationships among sentiments, contextual cues, and expressive modalities. Recent advances in DL have demonstrated strong performance in sentiment analysis, offering enhanced capabilities for mining, processing, and inferring intricate patterns across textual, visual, and multimodal data (Ahmed et al., 2024).

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https://doi.org/10.1016/j.neunet.2026.108885
Received 14 July 2025; Received in revised form 23 February 2026; Accepted 20 March 2026
Available online 21 March 2026

Recently developed graph-based methodologies have constructed heterogeneous sentiment networks to extract multi-perspective features from diverse data sources, based on the premise that similar sentiment expressions share common semantic characteristics. Autoencoder-based approaches, consisting of encoder–decoder architectures, have also been employed to learn compressed representations of sentiment data, preserving essential semantic information while reducing dimensionality. Despite these advancements, several critical challenges remain. Although DL methods demonstrate strong performance on sequential and multimodal data, they often struggle to handle complex graph-structured information. In particular, effectively modeling large-scale, heterogeneous sentiment data for robust representation learning remains challenging, especially when contextual information is sparse (Agbesi et al., 2025, 2024). Graph-based sentiment analysis is founded on the assumption that similar sentiment expressions are associated with comparable contextual and affective attributes. However, many existing studies focus primarily on bipartite networks (e.g., sentiment–context networks), thereby overlooking higher-order interactions and richer embeddings in more complex heterogeneous graph structures. While these approaches exploit potential correlations among sentiment embeddings, they often struggle to capture the subtle, latent features that fundamentally shape sentiment representations. Moreover, despite the progress achieved by deep neural and graph-based sentiment analysis models, most existing frameworks focus on syntactic or semantic dependencies in isolation and fail to capture cross-domain sentiment–emotion–context interactions. Current models lack heterogeneous meta-path reasoning mechanisms that explicitly construct cross-type relational patterns (e.g., sentiment → context → emotion), which are not represented in homogeneous sentence-level graphs (Li et al., 2021; Zhang et al., 2022). In addition, many approaches rely on simple concatenation or gating for feature fusion, rather than preserving part–whole relationships through more advanced, structured integration mechanisms. Compared with two-channel fusion strategies or knowledge graph augmentation methods, there remains a need for models that explicitly capture heterogeneous sentiment–emotion interdependencies while maintaining computational efficiency.

To address these limitations, this study introduces a two-fold multi-perspective network model that integrates sentiment-related and context-related multi-perspective networks with a sentiment-paired network, motivated by recent advances in interaction modeling and substructure learning. The proposed Two-Fold Multi-Perspective Heterogeneous Graph Neural Network (TFMPHGNN) incorporates three key conceptual innovations: (1) heterogeneous meta-path reasoning to explicitly construct cross-type relational patterns (e.g., sentiment → context → emotion); (2) capsule-based hierarchical fusion that preserves part–whole feature relationships through dynamic routing rather than simple concatenation or gating; and (3) VAE-driven denoised feature learning combined with multi-perspective sentiment–emotion pair (SEP) graphs, jointly modeling semantic similarity and structural dependencies through a triple-channel (topological, semantic, and collaborative) GCN pipeline. Sentiment-related networks are inherently complex structures comprising nodes that represent sentiments, contexts, expressions, and emotions, with edges encoding semantic, syntactic, or associative relationships. Within these networks, sentiment pairs are formed by combining sentiment and emotion nodes to form joint representations that effectively model potential interactions and associations among affective components. Multi-perspective graphs are constructed from complementary viewpoints—topological (structural), semantic (meaning-based), and collaborative (integrated)—to enable comprehensive feature learning across heterogeneous networks. By leveraging diverse relational information, the proposed framework enhances representational richness and improves the overall expressive capacity of sentiment analysis models.

TFMPHGNN integrates heterogeneous network learning for sentiment- and context-related features with multi-perspective sentiment-pair similarity network learning. The framework employs a graph autoencoder combined with the dynamic routing capabilities of a Capsule Network (CapsNet) to extract rich feature representations that capture diverse associative patterns embedded within the meta-paths of sentiment expressions and contextual elements. Concurrently, high-quality sentiment and context features generated through a variational autoencoder (VAE) are incorporated into the final node representations to enhance robustness and reduce noise. The model further constructs a semantic graph based on sentiment pairs and a collaborative graph that shares weights with the topological graph, enabling the learning of sentiment-pair embeddings through a graph convolutional network (GCN). Beyond conventional sentiment datasets, the framework supports the integration of sentiment embeddings derived from multiple associative sources, including user interactions, contextual signals, and multimodal inputs. Importantly, the architecture is designed to flexibly accommodate previously unseen sentiment expressions or contextual information during both training and inference phases. Additionally, a new sentiment corpus, VaKSent-2025, is developed by integrating data from Sentiment140, IMDb reviews, SemEval-2018, and the NRC Word-Emotion Lexicon. This corpus is designed to establish a comprehensive heterogeneous network that exploits semantic relationships among diverse sentiment expressions, contextual components, and subjective lexicons, thereby strengthening representation learning and improving overall model generalization. The main contributions of this study are as follows:

1. The study introduces TFMPHGNN, a novel two-fold, multi-perspective heterogeneous GNN that integrates sentiment-, context-, and sentiment-pair similarity networks for enhanced sentiment analysis. Unlike transformer-based models, our framework utilizes meta-path-based graph encoders, augmented with a capsule network and a VAE, to capture intricate, multi-source relationships within heterogeneous sentiment networks.
2. A multi-channel GCN is designed to analyze multi-perspective graphs—including topological, semantic, and collaborative features—thereby learning deep representations of sentiment pairs.
3. The study also developed a new sentiment corpus, VaKSent-2025, generated from diverse sentiment repositories, including Sentiment140, IMDb reviews, Sem-Eval-2018, and the NRC Word-Emotion-based datasets, to support the construction of a robust heterogeneous sentiment network.
4. The proposed TFMPHGNN achieves a 4.67% accuracy improvement over benchmark graph-based models, along with gains of 2.7% in F1-micro and 4.2% in F1-weighted, indicating robust and balanced performance across classes. Extended ablation analysis further confirms that these improvements represent substantive performance gains, demonstrating that the collaborative fusion channel consistently outperforms both topological-only and semantic-only variants, thereby confirming the synergistic benefit of integrating multiple graph perspectives.

The remainder of this paper is organized as follows: Section 2 presents a detailed review of related work on both deep neural networks and graph-based methods applied to sentiment analysis. Section 3 presents a detailed description of TFMPHGNN. Section 4 discusses the experimental setup and achieved results. The study concludes in Section 5.

2. Related work

This section provides an extensive review of related works in sentiment analysis relevant to the proposed TFMPHGNN. In particular, Section 2.1 presents reviews on CapsNets, Section 2.2 on VAE, and Section 2.3 on transformer-based sentiment analysis. Additionally, Section 2.4 reviews graph-based sentiment analysis models, including contrastive learning in Section 2.4.1 and a hybrid GCN-based model in Section 2.4.2.

2.1. Capsule networks

CapsNets, introduced by Sabour et al. (2017), represent an advanced neural architecture designed to address limitations of conventional convolutional neural networks (CNNs), particularly their inability to model hierarchical pose relationships among features. Unlike CNNs, which produce scalar neuron outputs, CapsNets employ vector-based representations known as capsules. In this framework, the length of a vector denotes the probability of a feature's presence, while its orientation encodes pose-related information such as position, orientation, and deformation. A key innovation of CapsNets is the dynamic routing mechanism, whereby lower-level capsules selectively route their outputs to higher-level capsules through an iterative agreement process. This routing-by-agreement strategy enables the accurate aggregation of lower-level components into coherent higher-level entities, making CapsNets particularly effective for tasks that require modeling spatial hierarchies and structured relationships (Hinton et al., 2018). In the context of sentiment analysis, this capability is especially valuable for capturing hierarchical linguistic structures, such as aspect–opinion pairs and dependency-based sentiment expressions, where conventional CNNs may overlook structural or sequential dependencies. Recent studies have explored the application of CapsNets to sentiment analysis with promising results. For instance, Zhao et al. (2018) integrated CapsNets with attention mechanisms to improve sentiment classification, demonstrating their effectiveness in preserving syntactic and semantic information. Similarly, Deng et al. (2021) proposed a capsule-attention network for aspect-level sentiment analysis, highlighting the model's ability to capture fine-grained opinion features. Furthermore, Shen et al. (2018) developed a directional self-attention capsule network that integrates capsule representations with directional self-attention for text encoding, enhancing sentiment and sentence classification through improved contextual modeling. These studies underscore the effectiveness of CapsNets in preserving hierarchical and compositional semantics, which are essential for complex and fine-grained sentiment classification tasks.

2.2. Variational autoencoders

Variational Autoencoders (VAEs), introduced by Kingma et al. (2013), are deep generative models designed to learn probabilistic latent-variable representations of input data. Unlike conventional autoencoders, which deterministically map inputs to a fixed latent space, VAEs encode inputs as distributions over latent variables—typically modeled as Gaussian distributions. This probabilistic formulation enables robust low-dimensional representation learning while also supporting data generation. A VAE consists of two primary components: an encoder that maps input data to a latent distribution, and a decoder that reconstructs data samples from this distribution. The training objective combines a reconstruction loss with a Kullback–Leibler (KL) divergence regularization term, which encourages the learned latent distribution to approximate a predefined prior, commonly a standard normal distribution. This composite loss ensures that the latent space is smooth, continuous, and well-structured, thereby improving generalization and stability in downstream tasks (Rezende et al., 2014). In sentiment analysis, VAEs have been employed to obtain robust embeddings that capture semantic variability while mitigating noise. For example, Wang et al. (2025) introduced a heterogeneous graph convolutional network (HGCN) that combines an improved VAE with a contrastive learning module. In their framework, dependency graphs are constructed from syntactic dependency trees to model explicit syntactic relationships. The VAE, augmented with a self-attention mechanism, extracts both global and local features, effectively capturing interactions between aspect terms and contextual words and addressing long-distance dependency challenges. Furthermore, a CapsNet-based contrastive learning objective is incorporated to enhance the model's ability to distinguish nuanced semantic and emotional representations across words. Similarly, Lu et al. (2020) employed a VAE to model latent sentiment attributes, improving classification robustness under noisy conditions. In addition, Fu et al. (2019) utilized VAEs for semi-supervised sentiment classification, leveraging unlabeled data to enhance predictive performance. Overall, VAEs are particularly effective for denoising high-dimensional sentiment features and generating compact, informative embeddings, making them well-suited for integration into graph-based and deep sentiment analysis frameworks.

2.3. Transformer-based models

The emergence of large-scale pretrained language models (PLMs) has introduced advanced mechanisms for contextual understanding and sentiment polarity detection, yielding strong performance across a wide range of downstream tasks (Agbesi et al., 2023, 2022). For instance, Hartvigsen et al. (2023) modified key hyperparameters of BERT to enhance performance; however, their approach resulted in overfitting. Similarly, Sun et al. (2023) applied embedding factorization and cross-layer pooling to reduce memory consumption and improve classification accuracy, although the performance gains were not consistent across tasks. Despite their effectiveness, PLM-based approaches present several challenges. Fine-tuning separate models for individual classification tasks is computationally expensive and may limit task-specific specialization (Rethmeier & Augenstein, 2023). Moreover, BERT-based sentiment analysis models can struggle with negation scope handling due to fixed context window constraints (Lin et al., 2020). DistilBERT was introduced as a compressed, general-purpose alternative to BERT through knowledge distillation; however, this reduction often comes at the cost of decreased performance (Dogra et al., 2021). In another effort, ReprBERT incorporated enhanced lexical and syntactic representations by extending the masking strategy during pretraining (Yao et al., 2022). Nevertheless, the highly parameterized and computationally intensive nature of these PLMs increases the risk of overfitting and reduces interpretability, particularly in multilingual and resource-constrained settings. To address these limitations, the present study adopts a graph-based framework to model complex word relationships more explicitly. By preserving critical semantic features while reducing dimensionality, the proposed approach enhances contextual integration, improves interpretability, and strengthens model generalization and robustness.

2.4. Graph-based sentiment analysis models

Recent studies highlight the growing importance of advanced graph-based models in sentiment analysis. For instance, Gao et al. (2024) demonstrated that GNNs effectively capture syntactic dependencies, which are essential given the critical role of sentence structure in determining sentiment. Their text classification optimization strategy aligns closely with efforts to integrate syntactic features into structured learning frameworks. In parallel, Yang et al. (2024a) explored the application of large language models (LLMs) to emotion analysis, showcasing their ability to manage nuanced linguistic phenomena and enhance sentiment extraction. Similarly, Cheng et al. (2024) leveraged ELMo embeddings and multimodal transformers to construct contextualized representations that incorporate both syntactic and semantic information, emphasizing the necessity of high-dimensional feature representations within GNN-based architectures. Optimization techniques have also received considerable attention. Ma et al. (2024) examined gradient descent strategies for training deep neural networks, including GNNs, and proposed practical methods for improving convergence and stability in complex architectures. In a related direction, Wang et al. (2024c) applied numerical-difference techniques to extract grammatical features, further strengthening the capacity of GNNs to model syntactic structures effectively. Although not exclusively focused on sentiment analysis, several studies provide valuable methodological insights for graph-based modeling. For example, Yang et al. (2024b) introduced a dynamic hypergraph prediction framework, conceptually related to modeling higher-order word interactions in textual data. Likewise, Gu et al. (2024)

Table 1
Summary of key strengths and limitations of recent graph-based methods.

Model	Strengths	Limitations
HM-GNN (Wu et al., 2024)	Integrates syntactic structure and positional information	Limited semantic modeling; underperforms in mixed contexts
HD-GCN (Zhou et al., 2023)	Hierarchical structure for syntactic-semantic fusion	High complexity; does not leverage emotion-specific information
SSEGCN (Zhang et al., 2022)	Employs syntactic mask and self-attention	Fails to incorporate multi-perspective node relations
IDGNN (Wang et al., 2024b)	Merges relational GAT and semantic GCN	Lacks support for collaborative graph learning or capsule routing
DualGCN (Li et al., 2021)	Combines syntactic and semantic features	Sensitive to graph sparsity; less robust under imbalance
TCKGCN (Hao et al., 2024)	Uses contextual semantics and SenticNet	Does not include hierarchical representation modeling (e.g., capsule fusion)
DGGCN (Liu et al., 2023)	Incorporates emotional signals via gating	Poor performance due to weak edge construction and limited perspectives
SSIN (Wang et al., 2024a)	Multi-source integration with syntactic attention	High model complexity; slow convergence and weaker interpretability

investigated spatio-temporal aggregation mechanisms in graph models, offering techniques that can be adapted to incorporate positional and contextual dependencies within GNN-based sentiment analysis frameworks.

2.4.1. *Contrastive learning*

Recent advances in GNNs have encouraged the widespread adoption of graph-based architectures, such as GCNs and GATs, for sentiment analysis tasks. For instance, (Zhang et al., 2019) constructed dependency graphs to represent sentence structures and applied GCNs to extract syntactic features for downstream sentiment classification. Similarly, Wang et al. (2020) proposed an aspect-oriented dependency model by refining traditional dependency trees and leveraging GATs to assign greater importance to context words closely related to specific aspects. In another study, Zhang and Qian (2020) explored the relationship between linguistic word co-occurrence and syntactic structure, introducing a hierarchical framework that integrates text-level co-occurrence information with syntactic dependency groups. Furthermore, Li et al. (2021) introduced a multi-GCN architecture that integrates multiple modalities to capture both syntactic dependencies and semantic representations more effectively. Likewise, Zhang et al. (2022) proposed a semantic–syntax enhanced GCN framework that incorporates attention mechanisms to improve contextual information modeling and strengthen feature representation. These studies demonstrate the effectiveness of GNN-based approaches in modeling structured linguistic information for sentiment analysis.

2.4.2. *Hybrid GCN-based*

Wang et al. (2023) proposed a dual-channel GNN architecture that integrates semantic and multi-granularity features by combining GAT with BERT representations. Similarly, Hao et al. (2024) introduced a knowledge-enhanced framework that incorporates ConceptNet-based conceptual knowledge, semantic metadata, and SenticNet sentiment memory to improve sentence-level aspect representation for downstream tasks. In another study, Zhou et al. (2023) developed a syntax-oriented graph model that integrates dual specialized GCNs for syntactic and semantic feature extraction. For semantic modeling, a self-attention matrix is employed to capture node-level correlations, while a dependency-feature-aware GCN is designed to process syntactic structures by adapting adjacency matrices to different dependency types. Their multi-layered architecture enables each layer to learn a distinct connection matrix, thereby enhancing the extraction of linguistic information at varying levels of abstraction. Similarly, Wang et al. (2024a) introduced SSIN, a model that jointly leverages semantic and syntactic features for sentiment analysis. Additionally, Yu et al. (2025) proposed a dual syntactic graph network combined with joint contrastive learning to further enhance downstream task performance. A detailed comparison of the strengths and limitations of these approaches is provided in Table 1. Although recent frameworks such as DualGCN, TCKGCN, and SSEGCN offer effective two- or three-channel fusion strategies for syntactic and semantic feature integration, they share three key limitations that the proposed TFMPHGNN addresses: (i) homogeneous graph assumptions, where models generally treat sentiment graphs as homogeneous structures, overlooking cross-type relational interactions such as sentiment–emotion–context meta-paths; (ii) simplistic feature fusion strategies, where many approaches rely on concatenation, gating, or single-head attention mechanisms, which do not adequately preserve hierarchical relationships among meta-path embeddings; (iii) limited robustness mechanisms, where most models directly propagate potentially noisy features through graph layers without incorporating dedicated denoising or latent regularization components. TFMPHGNN overcomes these challenges through a heterogeneous meta-path encoder that captures type-specific relational semantics, a capsule-based routing mechanism that adaptively reweights meta-path embeddings while preserving spatial hierarchies, and a VAE module that learns robust latent representations resistant to noise. These components enable deeper contextual modeling and improved generalization compared to prior approaches, as summarized in Table 2.

Table 2
Key distinctions of recent graph-based methods.

Model	Meta-Path Types	Fusion Mechanism	Denoising	Multi-Perspective SEP
DualGCN	Homogeneous	Concatenation	×	×
TCKGCN	External KB	Gating	×	×
SSEGCN	Homogeneous	Self-Attention	×	×
TFMPHGNN	Heterogeneous	Capsule Routing	✓ (VAE)	✓ (3-channel)

3. TFMPHGNN

The proposed TFMPHGNN framework integrates a meta-path-based graph encoder with a capsule neural network to model complex relationships within a heterogeneous network composed of sentiments, expressions, contexts, and emotions. The meta-path encoder captures type-specific relational semantics, while the capsule network preserves hierarchical and compositional structures through dynamic routing. Subsequently, TFMPHGNN employs a VAE to extract low-dimensional sentiment representations. The VAE learns a continuous latent space by encoding potentially noisy inputs and reconstructing denoised outputs, thereby ensuring that the model captures the underlying semantic structure of sentiment features while enhancing robustness. Following latent feature extraction, sentiment-pair feature vectors are constructed from the attributes of the corresponding sentiment nodes. For each sentiment pair, representation vectors are learned using GCNs applied to multi-perspective graphs, including topological (structural), semantic (meaning-based), and collaborative (integrated) graphs. Finally, TFMPHGNN employs a multi-layer fully connected network that takes the learned sentiment-pair representation vectors as input to predict associations between sentiments. This end-to-end architecture enables comprehensive relational modeling, robust feature learning, and effective sentiment association prediction within heterogeneous graph structures.

3.1. *Meta-path-based sentiment representation*

As illustrated in Fig. 1, the proposed framework leverages meta-paths to generate sentiment and emotion representations, producing low-dimensional embedding vectors that encapsulate rich relational information. The model extracts heterogeneous data from four

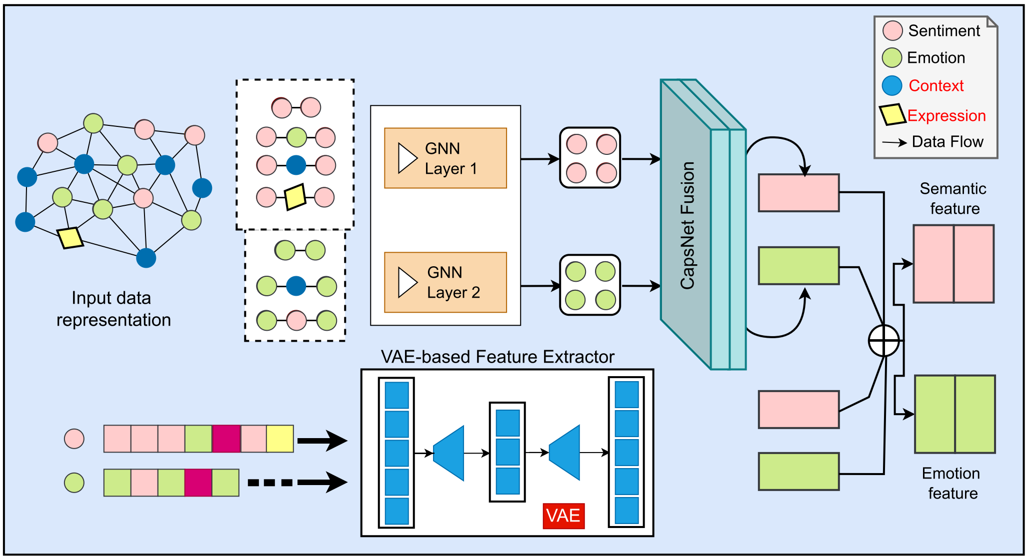


Fig. 1. The proposed TFMPHGNN framework. GNNs are used to extract multiple information associations in the meta-path module, and the VAE generates sentiment and emotion features.

sentiment-related networks and two emotion-related networks. In particular, the sentiment-related networks model polarity-based relationships among sentiments, expressions, and contexts, whereas the emotion-related networks capture affective attributes, thereby enriching the semantic representation of sentiment nodes. Six relational matrices are constructed as primary data sources: $D_{sentiment-sentiment}$, $D_{sentiment-context}$, $D_{sentiment-expression}$, $D_{sentiment-emotion}$, $D_{emotion-context}$, and $D_{emotion-emotion}$. Each matrix is binary, where an entry is assigned a value of 1 if a relationship exists between two nodes and 0 otherwise. For example, an element $D_{i,j}$ in $D_{\text{sentiment-context}}$ represents the relationship between node i (a sentiment) and node j (a context). Specifically, $D_{i,j} = 1$ indicates the presence of a relationship, while $D_{i,j} = 0$ denotes its absence. To construct the sentiment adjacency matrix, the matrix $D_{\text{sentiment-sentiment}}$, $D_{\text{sentiment-context}}$, $D_{\text{sentiment-expression}}$, and $D_{\text{sentiment-emotion}}$ are horizontally concatenated, such that each row corresponds to the descriptive feature profile of a specific sentiment node. This integration effectively combines four complementary relational dimensions into a unified representation, thereby enhancing the expressiveness and analytical capacity of sentiment modeling. Similarly, the emotion adjacency matrix is formed by concatenating $D_{\text{emotion-sentiment}}$, $D_{\text{emotion-context}}$, and $D_{\text{emotion-emotion}}$, enabling a comprehensive representation of affective relationships within the heterogeneous network.

3.1.1. Multi-source neighboring feature representation

To effectively integrate sparse sentiment information, we aggregate features from both sentiment and emotion networks into a unified matrix representation. Focusing on the sentiment domain, we first construct a composite feature structure comprising multiple relational matrices. The matrix $D_{\text{sentiment-sentiment}} \in \mathbb{R}^{M_s \times M_s}$ encodes pairwise relationships among sentiment nodes, where an entry is assigned a value of 1 if two sentiments are related and 0 otherwise; here M_s denotes the number of sentiment nodes. In addition, $D_{\text{sentiment-emotion}} \in \mathbb{R}^{M_s \times M_e}$ captures the interaction scores between sentiments and emotions, with M_e representing the number of emotion nodes. The ith row of this matrix corresponds to the sentiment–emotion interaction features of sentiment s_i. Similarly, $D_{\text{sentiment-context}} \in \mathbb{R}^{M_s \times M_c}$ and $D_{\text{sentiment-response}} \in \mathbb{R}^{M_s \times M_r}$ model the associations between sentiments and contexts, and sentiments and responses, respectively. An element, $(D_{\text{sentiment-context}})_{ij}$ for example, denotes the relationship score between sentiment s_i and context c_j (where M_c and M_r represent the numbers of context and response nodes. Collectively, the matrices $D_{\text{sentiment-sentiment}}$, $D_{\text{sentiment-emotion}}$, $D_{\text{sentiment-context}}$, and $D_{\text{sentiment-response}}$ serve as complementary feature matrices for sentiment representation. The ith row across these matrices provides a multi-perspective embedding of the sentiment node s_i. By concatenating these matrices horizontally, we obtain the characteristic sentiment matrix $D_G \in \mathbb{R}^{M_s \times (M_s + M_e + M_c + M_r)}$, where each row corresponds to the integrated feature embedding of a sentiment node s_i. In a similar manner, the emotion feature matrix is constructed by concatenating $D_{\text{sentiment-emotion}}$, $D_{\text{emotion-emotion}}$, and $D_{\text{emotion-context}}$, yielding $D_G \in \mathbb{R}^{M_e \times (M_s + M_e + M_c)}$. Finally, a variational autoencoder (VAE) is employed to transform these high-dimensional feature representations into compact, low-dimensional latent embeddings. By incorporating the VAE into TFMPHGNN, the model learns structured and expressive sentiment representations that are robust to noise. This probabilistic embedding process enhances generalization, improves stability, and mitigates overfitting compared to traditional deterministic autoencoder approaches.

The VAE is a deep generative framework designed to extract informative features while learning a probabilistic latent representation of high-dimensional input data (Cemgil et al., 2020). Unlike conventional autoencoders, which produce deterministic latent embeddings, the VAE encodes each input as a probability distribution by learning its parameters, typically the mean and variance—thereby capturing inherent uncertainty and variability in the data. The training objective of the VAE is defined by a composite loss function comprising two components: (i) a reconstruction loss (e.g., mean squared error) to ensure fidelity between the reconstructed output and the original input, and (ii) a Kullback–Leibler (KL) divergence term that regularizes the latent distribution by encouraging it to approximate a standard normal prior. To enhance numerical stability and convergence, the optimization process typically employs adaptive algorithms such as RMSProp, often combined with smooth activation functions such as softplus. Through this process, the VAE learns robust, low-dimensional embeddings for sentiment and emotion nodes, denoted as $X_{\text{sentiment}} \in \mathbb{R}^{N_s \times \text{Dim}}$ and $X_{\text{emotion}} \in \mathbb{R}^{N_e \times \text{Dim}}$ respectively. In our implementation, the latent dimensionality (Dim) is set to 64, as specified in Table 3.

3.1.2. Meta-based GCN module

Conventional GCN-based sentiment analysis methods often suffer from the loss of critical contextual information due to their fixed aggregation structures. To overcome this limitation, we introduce meta-paths to capture rich contextual dependencies within the heterogeneous sentiment graph θ (Hao et al., 2024). In particular, we derive distinct feature representations for sentiment and

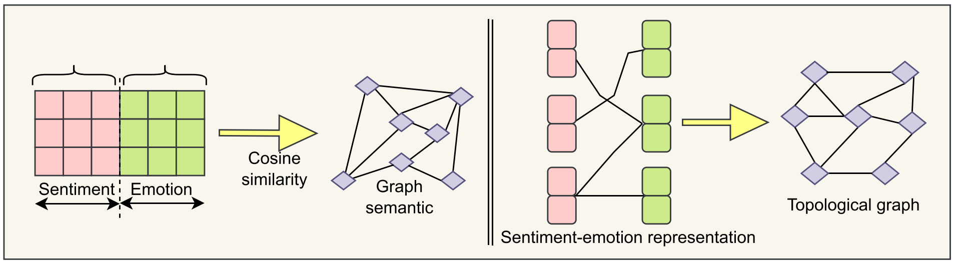


Fig. 2. Framework of the semantic–topological graph construction used for sentiment–emotion pair modeling.

emotion nodes through a set of predefined meta-paths. For sentiments, we consider four types of meta-paths: (i) sentiment–sentiment, (ii) sentiment–emotion–sentiment, (iii) sentiment–context–sentiment, and (iv) sentiment–expression–sentiment. For emotions, three meta-paths are constructed: (i) emotion–emotion, (ii) emotion–sentiment–emotion, and (iii) emotion–context–emotion. In heterogeneous sentiment networks, a meta-path is formally defined as an ordered sequence of node types connected via specific relation types, for example, $Y_1 \rightarrow R_1 \rightarrow Y_2 \rightarrow R_2 \rightarrow Y_3 \rightarrow R_3 \rightarrow, \dots, \rightarrow Y_n$, where Y_i denotes a node type and R_i represents the relation between node types Y_i and Y_{i+1}. Given the diversity of node categories and relation types in multi-typed heterogeneous sentiment graphs, constructing multiple meta-paths enables modeling complex semantic interdependencies. For example, the meta-path sentiment → context → sentiment indicates that two sentiment nodes are semantically connected through a shared contextual element. Similarly, emotion → context → emotion captures relationships between emotion nodes that co-occur within the same contextual setting. Let $Q = \{Q_1, Q_2, \dots, Q_D\}$ denote the set of constructed meta-paths for the heterogeneous sentiment graph θ, where D is the total number of meta-paths. Each meta-path Q_1 corresponds to an adjacency matrix Z_l forming the set $Z = \{Z_1, Z_2, \dots, Z_D\}$. For each meta-path Q_l, we employ a GCN to learn feature embeddings for sentiment and emotion nodes. Given a specific meta-path–induced adjacency matrix Z_l, the feature representation of node i at layer t is computed via a graph convolution operation. The activation of node i for feature dimension k in the layer t is expressed in Eq. (1) as follows:

$$h_{i,k}^{(t)} = \sigma\left(\sum_{j\in\mathcal{N}_i^{(l)}} \frac{1}{c_{ij}} h_{j,k}^{(t-1)} W_k^{(t)}\right), \tag{1}$$

where $\mathcal{N}_i^{(l)}$ denotes the set of neighbors of node i under meta-path Q_l, c_{ij} is a normalization coefficient (e.g., degree-based normalization), $W^{(t)}$ is the learnable weight matrix at layer t, and $\sigma(\cdot)$ represents a nonlinear activation function (ReLU in our implementation). Equivalently, in matrix form with self-loops incorporated, let $\widetilde{Z}_{p_l}$ denote the augmented adjacency matrix and $\widetilde{D}_{p_l}$ its corresponding degree matrix, where $\widetilde{D}_{p_l,i,i} = \sum_{j=1}^{n} \widetilde{Z}_{p_l,i,j}$ and n is the total number of nodes. This formulation ensures normalized neighborhood aggregation and stable feature propagation across layers. Since different meta-paths encode complementary semantic information, separate transformation parameters are learned for each meta-path. Let the sentiment embeddings obtained from different meta-paths be denoted as:

$$v_\pi = \{v_{\omega_1}, v_{\omega_2}, \dots, v_{\omega_{D_t}}\}, \tag{2}$$

where D_t is the number of meta-paths associated with sentiment nodes. To adaptively fuse these multi-view embeddings, we employ an attention mechanism to assign importance weights to each meta-path. The attention score for the i-th meta-path embedding is computed as:

$$Y_{\omega_i} = k \cdot \tanh\left(W v_{\omega_i} + b\right), \tag{3}$$

where k, W, and b are trainable parameters. The final aggregated sentiment representation is then obtained as:

$$h_{\text{sentiment}} = \sum_{i=1}^{D_t} \text{softmax}(\alpha_{\omega_i}) \cdot v_{\omega_i}. \tag{4}$$

By normalizing the attention coefficients α_{ω_i} via the softmax function, the model dynamically emphasizes the most informative meta-paths while suppressing less relevant ones. This meta-path–aware attention mechanism enables effective integration of heterogeneous semantic information, thereby yielding more expressive and discriminative sentiment representations.

3.2. Multi-perspective sentiment-Emotion pair network

To explore the intrinsic relationships between sentiments and emotions more comprehensively, we construct sentiment–emotion pairs (SEPs) by combining each sentiment with each emotion. Each pair is denoted as SEP_i. The embedding of the i-th SEP node is defined as

$$\rho_i^{\text{SEP}} = \left[\rho_s^{\text{sentiment}}\,;\,\rho_v^{\text{emotion}}\right],$$

where $\rho_s^{\text{sentiment}}$ and ρ_v^{emotion} represent the learned embeddings of the corresponding sentiment and emotion nodes, respectively, and $[\cdot;\cdot]$ denotes vector concatenation. This formulation produces a unified representation that jointly encodes sentiment and emotion characteristics. The embeddings of all SEP nodes are subsequently aggregated into a matrix $Z_n \in \mathbb{R}^{M_{\text{SEP}}\times d}$, where M_{SEP} denotes the total number of sentiment–emotion pairs and d represents the embedding dimensionality. Inspired by recent studies (Fu et al., 2016), we model SEPs in both topological and semantic spaces to capture complementary structural and contextual relationships. Accordingly, two distinct yet interrelated networks are constructed: (i) a topological network, which encodes structural dependencies among nodes, and (ii) a semantic network, which captures semantic similarities based on feature representations. The overall framework is illustrated in Fig. 2. Each network consists of two node types—sentiments and emotions—allowing the model to capture heterogeneous interactions from multiple perspectives. These networks form a multi-view heterogeneous graph that integrates structural and semantic information, thereby enabling more expressive modeling of sentiment–emotion relationships.

3.2.1. Topological graph construction

The topological graph, denoted as $\theta_n = (Z_t, \rho_{\text{SEP}})$, is constructed to model the structural relationships among sentiment–emotion pairs (SEPs). In this graph, an undirected edge is established between two SEP nodes if they share either a common sentiment or a common emotion component. The structural dependencies are encoded by an adjacency matrix $Z_t \in \mathbb{R}^{M_{\text{SEP}}\times M_{\text{SEP}}}$, where M_{SEP} denotes the total number of SEPs. The entry $(Z_t)_{ij} = 1$ indicates that the i-th and j-th SEP nodes share at least one common element (sentiment or emotion), while $(Z_t)_{ij} = 0$ otherwise. This construction enables the topological graph to explicitly capture structural correlations among SEPs, facilitating effective propagation of relational information in subsequent graph-based learning.

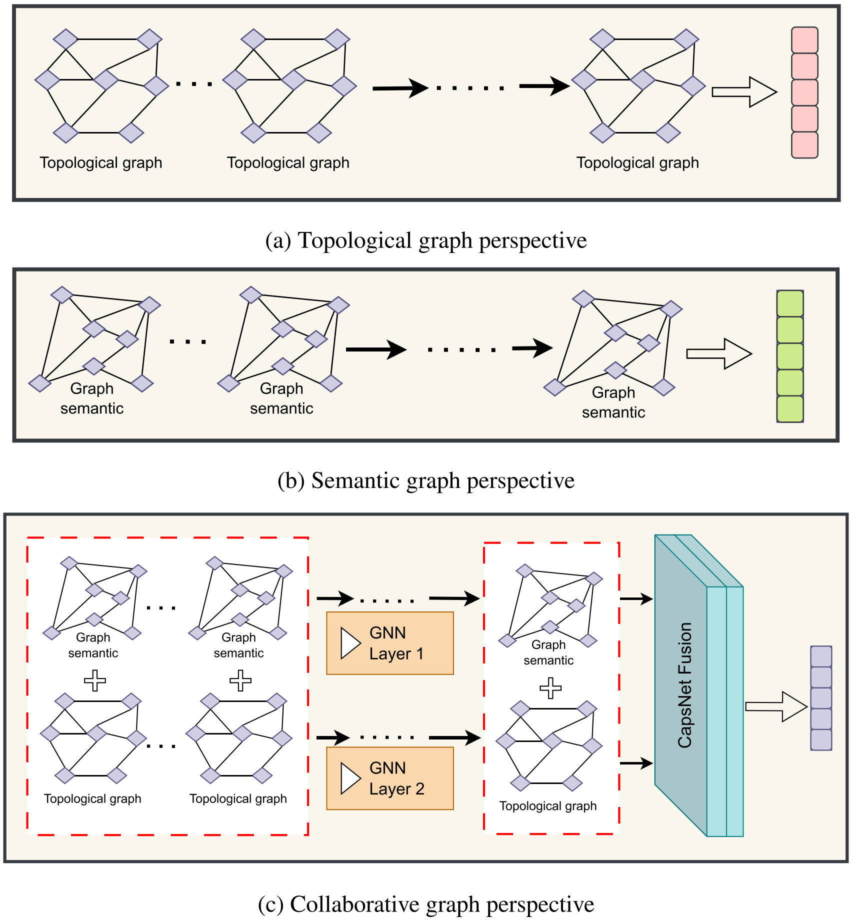


Fig. 3. Multi-perspective representation learning in TFMPHGNN. (a) Topological channel—captures structural relationships among sentiment–emotion pairs through shared nodes. (b) Semantic channel—models similarity-based associations derived from latent features. (c) Collaborative channel—integrates both topological and semantic embeddings through shared weights to form a unified representation. Arrows indicate data flow between layers, while capsules represent CapsNet-based fusion of multi-channel embeddings.

3.2.2. *Semantic graph construction*

In this section, we construct a semantic graph to model similarity-based relationships among sentiment–emotion pairs (SEPs). Unlike the topological graph, which captures structural dependencies, the semantic graph is built upon the similarity of SEP feature representations. In particular, we adopt a K-nearest neighbor (KNN) strategy (Kramer & Kramer, 2013). For each SEP node SEP_i, the cosine similarity between its embedding ρ_i^{SEP} and those of all other SEPs is computed. The top P most similar SEPs are then selected as the semantic neighbors of SEP_i. Based on this procedure, we construct the adjacency matrix $Z_s \in \mathbb{R}^{M_{\mathrm{SEP}} \times M_{\mathrm{SEP}}}$, where M_{SEP} denotes the total number of SEPs. An entry $(Z_s)_{ij}$ is defined as

$$(Z_s)_{ij} = \begin{cases} 1, & \text{if } \mathrm{SEP}_j \text{ is among the top } P \text{ nearest neighbors of } \mathrm{SEP}_i, \\ 0, & \text{otherwise.} \end{cases}$$

This construction yields the semantic graph $\theta_s = (Z_s, \rho_{\mathrm{SEP}})$, in which Z_s encodes similarity-driven semantic relationships among SEPs. By capturing feature-level affinities, the semantic graph complements the structural information encoded in the topological graph, enabling a more comprehensive representation of sentiment–emotion interactions.

3.2.3. *Triple-channel SEP network*

After constructing the topology graph $\theta_n = (a_n; \rho_{\mathrm{SEP}})$, and the semantic graph $\theta_s = (a_s; \rho_{\mathrm{SEP}})$ for SEPs, we employ a multi-perspective GNN to derive their respective feature representations, as shown in Fig. 3. For the topological graph channel shown in Fig. 3(a), each node's activation in the layer ℓ is computed as follows:

$$a_{t,i,k}^{(\ell)} = \max\left(0, \sum_{m=1}^{d^{(\ell-1)}} \left(\sum_{j=1}^{n} \frac{\widetilde{Z}_{t,i,j}}{\sqrt{\widetilde{D}_{t,i,i}\,\widetilde{D}_{t,j,j}}} \cdot a_{t,j,m}^{(\ell-1)}\right) W_{t,m,k}^{(\ell)}\right), \tag{5}$$

where $a_{t,i,k}^{(\ell)}$ denotes the activation of a node i for a feature k at a layer ℓ in the topological graph. $\widetilde{Z}_{t,i,j}$ is the (i,j)-th entry of the augmented adjacency matrix for the topological channel and $\widetilde{D}_{t,i,i}$ is its corresponding degree. $a_{t,j,m}^{(\ell-1)}$ is the feature m of the node j in the layer $\ell-1$, $W_{t,m,k}^{(\ell)}$ is the learnable weight from the feature m to k, and $\max(0,\cdot)$ denotes the ReLU activation. Similarly, the semantic graph channel, as depicted in Fig. 3(b), is computed as follows:

$$a_{s,i,k}^{(\ell)} = \max\left(0, \sum_{m=1}^{d^{(\ell-1)}} \left(\sum_{j=1}^{n} \frac{\widetilde{Z}_{s,i,j}}{\sqrt{\widetilde{D}_{s,i,i}\,\widetilde{D}_{s,j,j}}} \cdot a_{s,j,m}^{(\ell-1)}\right) W_{s,m,k}^{(\ell)}\right), \tag{6}$$

where $a_{s,i,k}^{(\ell)}$ is the activation of the node i for the feature k at layer ℓ in the semantic graph. $\widetilde{Z}_s$ and $\widetilde{D}_s$ are the augmented adjacency matrix and its degree matrix, respectively, for the semantic channel. Finally, to learn cooperative representations, we introduce a third GCN with shared weights. However, in this triple-channel framework, the final cooperative embedding of each node often arises by combining the topological representation z_t and the semantic representation z_s (see Fig. 3(c)). Concretely, after obtaining the layer-wise embeddings$\{a_t^{(\ell)}, a_s^{(\ell)}\}$, the final cooperative representation z_c is formulated as follows:

$$a_c = \frac{a_n + a_s}{2}. \tag{7}$$

In this part, a_n and a_s denote the aggregated embeddings for all nodes in the topological and semantic channels, respectively. This triple-channel approach—topological, semantic, and cooperative—enhances the overall SEP representation by capturing structural information, semantic relationships, and cooperative features in a unified manner.

3.2.4. Complementarity of multi-perspective representations

The three perspectives address orthogonal aspects of the sentiment–emotion relationship. The topological perspective (Section 3.2.1) captures structural connectivity through shared nodes—for instance, SEP_1 (angry, complaint) and SEP_2 (angry, feedback) are linked via the common emotion "angry," modeling co-occurrence patterns and affective components regardless of contextual differences. The semantic perspective (Section 3.2.2) captures latent similarity through VAE-encoded representations—SEP_3 (frustrated, delay) and SEP_4 (annoyed, waiting) may lack shared nodes yet exhibit high cosine similarity, indicating semantic equivalence despite differing surface forms. The collaborative perspective (Section 3.2.3) fuses structural and semantic signals via shared weights as illustrated in Eq. (7). This fusion enables the model to identify reinforcing signals (structurally connected and semantically similar SEPs) versus conflicting signals (structurally distant yet semantically proximate pairs). By distinguishing between these signal types, the model effectively addresses cold-start problems for rare SEPs through information borrowing from semantically similar neighbors. For example, consider three SEPs: SEP_A (disappointed, service), SEP_B (disappointed, quality), and SEP_C (frustrated, service). The topological channel links $\text{SEP}_A \leftrightarrow \text{SEP}_B$ (shared emotion) and $\text{SEP}_A \leftrightarrow \text{SEP}_C$ (shared context). The semantic channel identifies $\text{SEP}_B \leftrightarrow \text{SEP}_C$ similarity (both encode negative product-related sentiments). The collaborative channel integrates both signals, inferring that SEP_C inherits properties from SEP_A (structural connection) and SEP_B (semantic similarity), thereby improving predictions for underrepresented SEPs through multi-perspective information fusion.

3.3. Sentiment prediction module

As described in Section 3.2.3, the topological, semantic, and integration feature embeddings of SEP nodes were derived. Subsequently, these diverse representations are fused using a capsule network rather than a transformer or attention mechanism. Let a_i^n, a_i^s, and a_i^c denote the three feature embeddings of node i. The capsule network processes these embeddings to produce the capsule outputs, v_i^n, v_i^s, and v_i^c, which reside in a higher-dimensional capsule space.

3.3.1. Capsule network operation

For each embedding $\{a_i^n, a_i^s, a_i^c\}$, the network first computes capsule activations $\{u_i^n, u_i^s, u_i^c\}$ via a transformation matrix $W \in \mathbb{R}^{d_h \times d_c}$, where d_h represents the embedding dimension of the SEP node in the final layer and d_c represents the capsule space dimensionality, and is represented as follows:

$$u_i^t = \text{squash}(W^t a_i^n + b^n), \tag{8}$$

$$u_i^s = \text{squash}(W^s a_i^s + b^s), \tag{9}$$

$$u_i^c = \text{squash}(W^c a_i^c + b^c), \tag{10}$$

where W^n, W^s, and W^c are learnable transformation matrices, b^n, b^s, and b^c are learnable biases, and squash$(\cdot)$ is a non-linear activation ensuring the vector length reflects the probability of each feature's presence. Next, we activate a dynamic routing algorithm that then iteratively adjusts coupling coefficients c_{ij} to determine the extent to which each capsule contributes to the final output. These coefficients are computed using a softmax over the logits b_{ij}, which are initialized to 0 and updated over successive routing iterations as follows:

$$c_{ij} = \frac{\exp(b_{ij})}{\sum_k \exp(b_{ik})}. \tag{11}$$

The final capsule outputs are derived as weighted sums of the transformed embeddings:

$$v_i^n = \sum_j c_{ij}\, u_j^n, \tag{12}$$

$$v_i^s = \sum_j c_{ij}\, u_j^s, \tag{13}$$

$$v_i^c = \sum_j c_{ij}\, u_j^c. \tag{14}$$

At this stage, v_i^n, v_i^s, and v_i^c represent the fused feature representations in the capsule space, capturing both the presence and spatial relationships of the features. In our experiments, we used Sabour-style dynamic routing with (K = 3) routing iterations, empirically selected as a trade-off between convergence and computation. Each lower-level capsule (u_{*j}) has dimension ($d_c = 16$), and the higher-level capsule count (N) was set to match output label groups (here, $N = 3$ for positive/negative/neutral SEP outputs). The logits (b_{ij}) were initialized to zero and updated via standard routing-by-agreement. We used the squash nonlinearity as in Sabour et al. (2017), and capsule weights (W) were learned with the same optimizer and learning rate as the rest of the network ($RAdam, lr = (1 \times 10^{-4})$, weight decay (1×10^{-8}), cf. Table 3. We also experimented with EM-routing (Hinton et al., 2018) and found similar trends but higher computational costs; the study reports results using the dynamic routing strategy with 3 iterations.

3.3.2. Final representation

To obtain the final representation of the node i for the sentiment analysis task, these capsule outputs are concatenated as follows:

$$h_i = \text{Concat}\left(v_i^n, v_i^s, v_i^c\right), \tag{15}$$

where v_i^n, v_i^s, v_i^c are the computed capsule outputs for topological, semantic, and integration features. Leveraging the capsule network's ability to model hierarchical relationships and spatial invariance yields a more robust and interpretable feature-fusion mechanism than transformer- or attention-based approaches. Finally, this aggregated representation is fed into a fully connected network to predict the sentiment score as follows:

$$\hat{y}_i = \phi\big(W_f\, h_i + b_f\big), \tag{16}$$

where $W_f \in \mathbb{R}^{d_o \times d_h}$ is the weight matrix, $b_f \in \mathbb{R}^{d_o}$ is the bias vector, d_h is the dimension of the concatenated feature, h_i, d_o is the output dimension, and $\phi(\cdot)$ denotes the softmax activation function.

4. Experiments, results and discussions

This section presents the VaKSent-2025 dataset in Section 4.1, provides a detailed description of the experimental setting in Section 4.2, and finally discusses the achieved results and their interpretations in Sections 4.4 to 4.6.

4.1. Vaksent-2025 dataset

The VaKSent-2025 dataset is a comprehensive heterogeneous sentiment analysis corpus that integrates five established resources into a unified sentiment–emotion–context–expression network. Conventional sentiment datasets, such as Sentiment140 and IMDb, provide extensive sentiment–context associations but lack explicit emotion and expression annotations, which are crucial for modeling deeper affective dependencies. In contrast, established emotion lexicons, including NRC and SenticNet, offer fine-grained emotional categorizations but contain limited contextual information, thereby restricting their effectiveness in

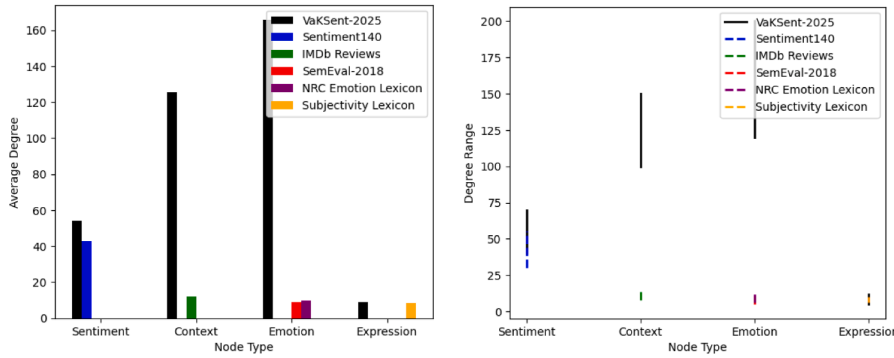


Fig. 4. Comparing the average degree of sentiment, context, emotion, and expression in the heterogeneous network constructed with VaKSent-2025 and other datasets.

multi-perspective learning frameworks. To overcome these limitations, VaKSent-2025 consolidates complementary data sources to jointly capture affective, semantic, and contextual relationships within a single heterogeneous graph structure. Sentiment nodes are primarily derived from Sentiment140[1], while context nodes are extracted from IMDb reviews[2]. The SemEval-2018 Affective Text dataset[3] contributes emotion annotations and sentiment polarity labels, thereby enriching sentiment–context associations. Additional sentiment–emotion correlations are incorporated from the NRC Word–Emotion Association Lexicon[4], and sentiment–expression relationships are obtained from the Subjectivity Lexicon[5]. In total, the constructed network comprises 5000 sentiment nodes and 4500 context nodes (sourced from Sentiment140 and IMDb), 3200 emotion nodes (from SemEval-2018 and NRC), and 2800 expression nodes (from the Subjectivity Lexicon). The emotion nodes are categorized into eight primary affective classes: joy, anger, sadness, fear, trust, disgust, surprise, and anticipation. To model the heterogeneous interactions among these entities, six distinct edge types are defined: sentiment–context, sentiment–sentiment, emotion–emotion, sentiment–emotion, sentiment–expression, and emotion–context relationships. The overall architecture of the VaKSent-2025 heterogeneous network is illustrated in Fig. 4. To ensure fairness and robustness, several bias mitigation strategies were implemented during dataset construction. Domain diversity was achieved by integrating short-form social media texts (Sentiment140), long-form reviews (IMDb), and expert emotion annotations (SemEval-2018 and NRC), thereby reducing platform-specific biases such as those stemming from Twitter's brevity or IMDb's product focus. Emotion balancing was enforced by maintaining at least 300 instances per NRC emotion class, preventing overrepresentation of frequent emotions such as anger and joy. Cross-source validation enhanced reliability by retaining only sentiment–emotion associations corroborated by at least two independent sources (e.g., SemEval and NRC), while noise filtering excluded low-confidence associations (similarity score < 0.3) and isolated nodes with fewer than two connections, minimizing weak or spurious relationships. The final heterogeneous network consists of 15,500 nodes and approximately 124,000 weighted edges, substantially exceeding the scale of previously utilized sentiment analysis datasets. As shown in Fig. 4, VaKSent-2025 achieves balanced average node degrees across all types, indicating successful integration without topological bias toward any particular data source or modality. This balance ensures equitable contribution of each domain to the overall network structure, thereby supporting robust, generalizable model learning and providing a comprehensive benchmark for multi-perspective sentiment and emotion analysis.

[1] https://www.kaggle.com/datasets/kazanova/sentiment140
[2] https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews
[3] https://www.kaggle.com/datasets/context/semeval-2018-task-ec
[4] https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm
[5] https://www.cs.cornell.edu/people/pabo/movie-review-data/

Table 3
Parameter Settings.

Hyperparameter	Value
Dim-VAE	64
1st GCN layer	128
2nd GCN layer	64
Optimizer	RAdam
Learning rate	1×10^{-4}
Weight decay rate (WDR)	1×10^{-8}
Dropout rate	0.3
Temperature parameter τ	0.6

4.2. Experimental setup

To ensure methodological rigor and prevent potential data leakage across the multi-source heterogeneous network, a total of 15,000 SEPs were generated from 5000 sentiment and 3200 emotion nodes, selecting only pairs that shared at least one context or expression node. We later employed a stratified splitting strategy to preserve the proportional distribution of data across key dimensions, namely: (i) sentiment polarity (positive, negative, and neutral), (ii) emotion category (eight NRC emotion classes), and (iii) source dataset (Sentiment140, IMDb, SemEval-2018, and NRC Word–Emotion). This approach ensured that each fold maintained a balanced representation of both affective and contextual diversity. A 5-fold cross-validation protocol was implemented, with each fold comprising 20% of the SEPs for testing and the remaining 80% for training. To further minimize indirect information leakage, node-level isolation was enforced: sentiment and emotion nodes involved in test SEPs were masked during meta-path construction in the training phase, preventing structural graph information from test nodes from propagating through shared edges. For Sentiment140, a chronologically ordered split was additionally applied, where training samples were drawn from 2008–2016 and test samples from 2017–2018, thereby preserving temporal consistency and preventing retrospective bias. To confirm the integrity of the data partitions, a leakage-prevention validation step was performed. This verified that no test SEP shared an identical (sentiment, context, emotion) tuple with any training sample. Moreover, the average cosine similarity between the train and test node embeddings was 0.23, indicating low overlap and sufficient representational separation between the partitions. As summarized in Table 3, optimal hyperparameter settings included a VAE latent dimension of 64, two GCN layers, a learning rate of 1×10^{-4}, and a dropout rate of 0.3, providing an effective balance between model expressiveness and regularization. All experiments were conducted on a single RTX 3050 Ti GPU server and implemented using the scikit-learn and PyTorch libraries.

4.3. Benchmark model

Eight recently designed models were selected for experimental comparison based on their performance, and each was implemented using the initial hyperparameter settings described in its original study.

Table 4
Average numerical scores for the baselines and the proposed framework. NOTE: All baseline methods were implemented with their initial hyperparameters.

Model	Accuracy	F1-macro	F1-micro	F1-weighted
SSEGCN (Zhang et al., 2022)	0.8613	0.8499	0.8601	0.8549
IDGNN (Wang et al., 2024b)	0.7954	0.7811	0.7889	0.7849
DGGCN (Liu et al., 2023)	0.6323	0.6042	0.6177	0.6215
DualGCN (Li et al., 2021)	0.8018	0.8003	0.8197	0.8099
HM-GNN (Wu et al., 2024)	0.5915	0.5822	0.5737	0.5816
TCKGCN (Hao et al., 2024)	0.8920	0.8755	0.9038	0.8888
HD-GCN (Zhou et al., 2023)	0.8496	0.8399	0.8382	0.8391
SSIN (Wang et al., 2024a)	0.7275	0.7023	0.7165	0.7200
TFMPHGNN (Ours)	**0.9387**	**0.9296**	**0.9311**	**0.9303**

- HM-GNN (Wu et al., 2024) is introduced to capture the complex relationships among sentence components by integrating syntactic structure and positional information.
- HD-GCN (Zhou et al., 2023) designed both syntactic and semantic information. It is structured as a multi-layered architecture to capture linguistic features at various levels, thereby enabling in-depth collaborative learning for sentiment analysis.
- SSEGCN (Zhang et al., 2022) introduced an attention mechanism that employs self-attention alongside a syntactic mask-matrix for downstream tasks.
- IDGNN (Wang et al., 2024b) combined syntactic and semantic features by using a relational GAT to get syntactic data from dependency trees and a semantic GCN to get semantic data from self-attention matrices. The model then fuses these features via two gate-based modules to generate the final representation.
- DualGCN (Li et al., 2021) simultaneously addressed the complementarity and intercorrelations between syntactic and semantic information.
- TCKGCN (Hao et al., 2024) examined the interconnections between contextual semantics, concepts, emotional dimensions, and the syntactic structure of a sentence for sentiment analysis via a three-channel fusion of information.
- DGGCN (Liu et al., 2023) integrated a GCN in a gating network, enhancing the emphasis on some aspects by amalgamating characteristics from several nodes. Furthermore, the graph network incorporates contextual emotional information.
- SSIN (Wang et al., 2024a) merged syntactic and semantic features to enhance knowledge acquisition. Its core network leverages isomorphism and syntax, while a semantically guided syntactic attention module refines the integrated semiotic representations.

4.4. Validating TFMPHGNN

The performance of TFMPHGNN is evaluated using accuracy, macro-F1, F1-micro, and F1-weighted metrics, as presented in Table 4, to facilitate a comprehensive comparison with other graph-based benchmark models. The best results are bolded, while the second best are underlined. As shown in the table, TFMPHGNN consistently outperforms all baseline models across all metrics, demonstrating its ability to model complex relationships in heterogeneous networks. This superior performance is attributed to the integration of meta-path-based graph encoders with capsule networks, which adeptly capture nuanced interactions involving sentiments, expressions, contexts, and emotions. The dynamic weighting of different meta-paths enabled by the capsule network results in a significant improvement over models such as DGGCN and HM-GNN, which do not employ such techniques. Although TCKGCN exhibits commendable performance by incorporating temporal context, achieving an accuracy of 0.8920, the 4.67% increase in accuracy with TFMPHGNN highlights the additional benefits provided by meta-path encoding and capsule networks, extending beyond temporal awareness. Moreover, the inferior performance of standalone models, such as DGGCN and HM-GNN, which lack these dynamic approaches, highlights

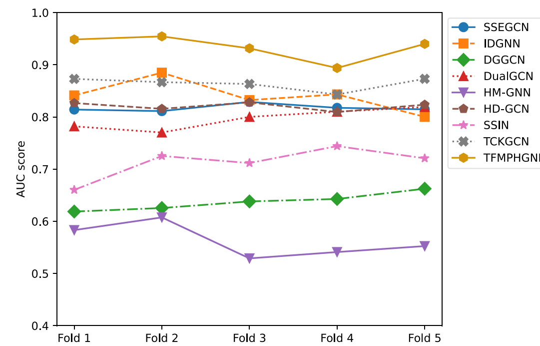


Fig. 5. Five-fold AUC scores for each model.

the need for advanced methods to model complex network structures effectively. This conclusion is further reinforced by the AUC analysis shown in Fig. 5, which confirms the robustness and generalizability of TFMPHGNN through its consistent performance across all folds. In all, the experimental results suggest that TFMPHGNN not only sets a new benchmark for heterogeneous graph-based tasks involving multifaceted relationships but also holds significant promise for real-world applications such as emotion-aware recommendation systems, sentiment analysis, and contextualized user modeling.

4.4.1. Statistical significance test

To ensure that the performance gains observed for TFMPHGNN were not due to random variation, we conducted statistical significance testing. Specifically, paired t-tests were performed between TFMPHGNN and the two strongest baseline models—TCKGCN and DualGCN—using results obtained from 5-fold cross-validation. The tests evaluated whether the improvements in F1-micro and accuracy were statistically significant at the 0.05 significance level. The results demonstrate that TFMPHGNN consistently outperforms both baselines. When compared with TCKGCN, the model achieved p-values of 0.0043 for F1-micro and 0.0038 for accuracy. Similarly, in comparison with DualGCN, the p-values were 0.0019 for F1-micro and 0.0024 for accuracy. Since all p-values are well below the 0.05 threshold, the improvements achieved by TFMPHGNN are statistically significant. These findings confirm that the observed performance gains are robust and unlikely to be attributable to chance, thereby reinforcing the effectiveness of the proposed model.

4.4.2. Feature relevance analysis

To gain deeper insight into the interpretability of TFMPHGNN, we conducted a feature importance analysis using SHAP values on the VaKSent-2025 dataset. The analysis reveals that contextual features play a critical role in shaping the model's predictions. For instance, context nodes such as "product defect" and "delivery delay" exerted the strongest influence on negative sentiment classifications, indicating that the model effectively captures situational cues associated with dissatisfaction. Conversely, emotion nodes including "joy" and "trust" were the most influential contributors in positive sentiment predictions, reflecting the model's sensitivity to affective signals. Expression nodes such as "not bad" and "too slow" were particularly important for fine-grained distinctions, helping the model differentiate between neutral and polarized sentiments. These findings demonstrate that TFMPHGNN does not rely solely on surface-level lexical patterns. Instead, it learns meaningful, domain-relevant associations across contextual, emotional, and expressive dimensions. This interpretability analysis further validates the effectiveness of the model's multi-source integration strategy and heterogeneous graph structure.

4.4.3. Scalability and computational analysis

As shown in Fig. 6, TFMPHGNN has a moderate training cost and the fastest inference time, despite its complexity. DualGCN and TCKGCN have similar training times but much slower inference due to less

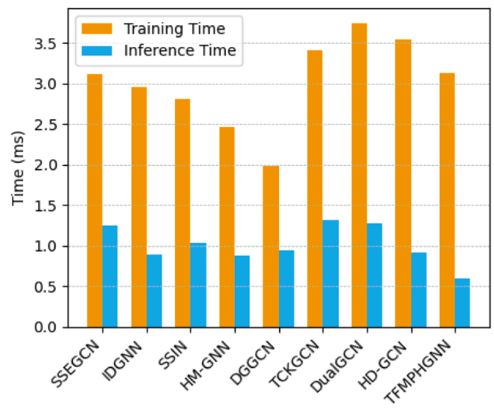


Fig. 6. Runtime comparison across models on VaKSent-2025.

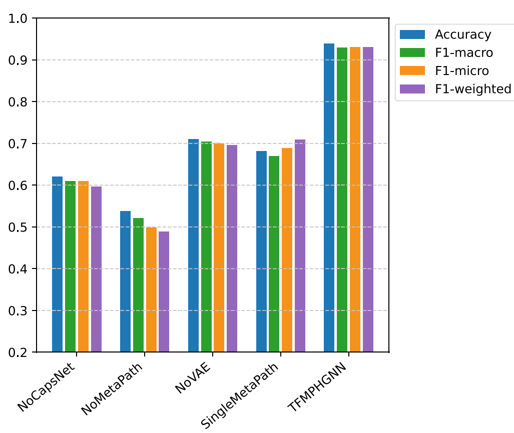


Fig. 7. A comparison of experimental results for TFMPHGNN and its variations.

efficient fusion strategies. DGGCN trains the fastest but achieves lower accuracy, highlighting a classic speed–accuracy trade-off. The effects of scaling by feature dimension and label count were systematically evaluated to understand the performance-efficiency trade-offs in the proposed model. As shown in Fig. 9(a), model performance peaks at a feature dimension of $d = 64$, beyond which training time increases significantly—proportional to d^2—while accuracy either stagnates or declines, likely due to overfitting. Therefore, the optimal dimensional range lies between 64 and 128, offering a balance between computational efficiency and classification performance. In terms of label scaling, TFMPHGNN effectively accommodates multi-class and fine-grained sentiment categories, including neutral, positive, negative, and mixed labels. Unlike traditional softmax-based models, using a CapsNet enables more efficient handling of label diversity by capturing spatial relationships among features, thereby mitigating parameter explosion. TFMPHGNN offers a strong balance between accuracy, convergence speed, and inference efficiency, making it well-suited for real-time sentiment applications. While not the fastest in training, it achieves fast inference ($<$ 1 ms) and efficiently handles high-dimensional inputs through its VAE component, enabling linear scalability with added features. Its modular design supports a range of input types and label complexities, enabling deployment in low-latency environments such as web apps and SME systems, and consistently outperforms baselines in inference speed.

4.5. Asymptotic computational complexity analysis

The overall time complexity of TFMPHGNN comprises four main computational components. First, the meta-path-based GCN encoding performs neighbor aggregation and feature transformation for each predefined meta-path, with a complexity of $O(D \cdot |V|^2)$, where D is the number of meta-paths and $|V|$ is the number of nodes. Although this is quadratic in the worst case, practical graphs are typically sparse, significantly reducing the actual computational cost. Second, the VAE compresses high-dimensional node attributes into low-dimensional latent vectors, with a forward pass complexity of $O(|V| \cdot d^2)$, where d is the latent dimension. The loss computation remains linear in the number of nodes. Third, triple-channel GCN fusion, which processes topological, semantic, and collaborative features per sentiment-emotion pair, has a per-layer complexity of $O(L \cdot |E| \cdot d^2)$, where L is the number of GCN layers and $|E|$ is the number of edges. Finally, the CapsNet used for representation fusion incurs a complexity of $O(K \cdot N^2 \cdot d^2)$, where N is the number of capsules, d is the capsule dimension, and K is the number of routing iterations. TFMPHGNN scales linearly with the number of nodes and labels in sparse settings and quadratically with the latent feature size during capsule fusion. TFMPHGNN achieves a balance between expressive modeling and computational tractability, ensuring scalability to mid- and large-scale sentiment analysis tasks.

4.6. Ablation experiment

An ablation experiment is conducted to assess the effectiveness of different components of TFMPHGNN, thereby confirming the beneficial contributions of each component to the sentiment analysis task. In this study, the variants of the proposed TFMPHGNN are as follows:

- NoCapsNet: Remove the capsule network from the meta-path-based graph encoder; instead, deploy a max pooling strategy to aggregate information across meta-paths.
- NoMetaPath: Remove the meta-path-based graph encoder entirely. Instead, treat the heterogeneous graph as a homogeneous graph and apply standard GNNs.
- NoVAE: Remove the VAE module, and only the meta-path learning module is used to obtain the features.
- SingleMetaPath: Use only one metapath instead of multiple metapaths.

The results of each variant are shown in Fig. 7. The figure shows that each component contributes significantly to TFMPHGNN's performance. For instance, when the capsule network from the meta-path-based graph encoder is removed (NoCapsNet), it fails to capture the detailed importance of different meta-paths, resulting in a noticeable decrease in performance, with accuracy, F1-macro, and F1-weighted dropping by 31.74%, 32.97%, and 33.40%, respectively. This severe performance degradation is due to the collapse of meta-path diversity, the destruction of hierarchical features, and the failure of the multi-perspective fusion model. This confirms that CapsNet is crucial for dynamically weighting meta-paths and improving classification performance. The NoMetaPath variant shows the largest performance decline, indicating that the meta-path-based graph encoder is crucial for capturing specific interaction patterns in heterogeneous networks. Without meta-paths, the model loses the ability to differentiate between various types of relationships (e.g., "User → Post" or "User → Comment"), resulting in unsatisfactory performance with accuracy and F1-macro scores of 0.5376 and 0.5211. This highlights the need to explicitly model diverse relationships via meta-paths. Removing the VAE module results in a moderate drop in performance relative to the proposed model, indicating that the VAE plays a significant role in feature representation. The VAE helps denoise and refine node features, thereby improving the quality of embeddings for downstream tasks. However, the meta-path learning module alone still provides useful representations, as indicated by higher scores than other ablated variants. Finally, using a single meta-path results in a notable decline in performance, though not as severe as removing the meta-path encoder entirely. This indicates that while a single meta-path can capture some relevant relationships, it is insufficient to model the full complexity of the heterogeneous network. Overall, Capsule dynamic routing enhances TFMPHGNN's performance by enabling

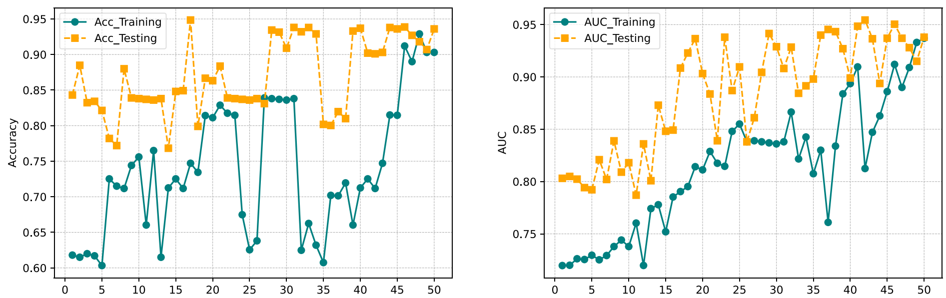
Table 5
Case studies comparing the truth labels of TFMPHGNN and baseline models.

Sentence under consideration	IDGNN	TCKGCN	DualGCN	HM-GNN	TFMPHGNN
A: The [pizza = neg] crust here is always burnt, but the [service = pos] is quite polite.	[neu] = no [pos] = yes	[neg] = yes [pos] = yes	[neg] = yes [neu] = no	[neu] = no [neu] = no	[neg] = yes [pos] = yes
B: Consider trying the [vitamin pills = pos] without a [prescription = neu].	[pos] = yes [neg] = no	[pos] = yes [neg] = no	[pos] = yes [neg] = no	[neg] = no [neg] = no	[pos] = yes [neu] = yes
C: The girls here are too young to understand what they've gotten themselves into when it comes to [Chinese cuisine = pos] and [pepper = pos].	[pos] = yes [neu] = no	[neg] = no [neu] = no	[pos] = yes [neg] = no	[neg] = no [neu] = no	[pos] = yes [pos] = yes
D: [The good thing = pos] is that there are no exams in the next two days, [haha! = pos]	[pos] = yes [neg] = no	[pos] = yes [pos] = yes	[neu] = no [pos] = yes	[neg] = no [pos] = yes	[pos] = yes [neu] = no
E: I got that [phone = neu] over 3 months ago and found no problems with the IOS or [updates = neu].	[pos] = no [pos] = no	[neu] = yes [pos] = no	[pos] = no [neg] = no	[pos] = no [pos] = no	[neu] = yes [neu] = yes
F: So, I'm not a big fan of [Boll's work = neg], but then [again, not = neg] many are.	[neu] = no [pos] = no	[pos] = no [pos] = no	[neu] = no [neg] = yes	[pos] = no [neg] = yes	[neg] = yes [neg] = yes
G: We see the evil mad scientist Dr. Krieger, played by Udo Kier, who makes genetically [modified = neu] soldiers, or GMS, as they are called.	[neu] = yes	[pos] = no	[pos] = no	[neg] = no	[neu] = yes
H: [Feeling alone = neg] on the set, Mr Boll invites three of his country-men to [play = pos] with him.	[neg] = yes [pos] = yes	[neu] = no [pos] = yes	[neu] = no [neu] = no	[pos] = no [pos] = yes	[neg] = yes [pos] = yes
I: Oh! Haha... dude, I don't really look at them unless someone says HEY, I ADDED YOU. Sorry, I'm so terrible at that. I need a [pop-up = neu]!	[neu] = yes	[neu] = yes	[pos] = no	[neg] = no	[pos] = no
J: This week just [seems to get longer = neg] and longer in terms of how much I need to do and how much I'm actually going to get [done = pos].	[neg] = yes [pos] = yes	[neg] = yes [pos] = yes	[neu] = no [pos] = yes	[neg] = yes [neu] = no	[neg] = yes [pos] = yes

Table 6
Results comparison of different methods.

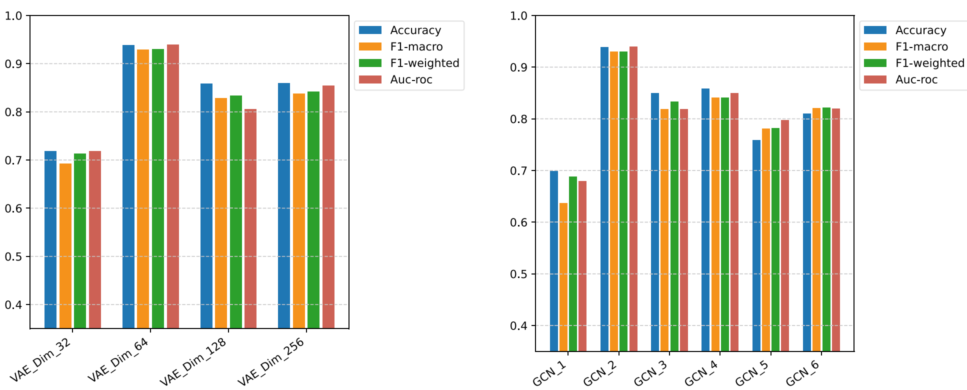
Model	Twitter		Lap14		Rest16		VAKSent-2025	
	Accuracy	F1-Macro	Accuracy	F1-Macro	Accuracy	F1-Macro	Accuracy	F1-Macro
ASGCN	0.7215	0.7040	0.7555	0.7105	0.8809	0.6748	–	–
DualGCN	0.7592	0.7429	0.7848	0.7474	–	–	0.8018	0.8003
MGAN	0.7254	0.7081	0.7527	0.7081	–	–	–	–
GLGCN	0.7326	0.7126	0.7691	0.7276	0.8847	0.6964	–	–
HD-GCN	0.7637	0.7467	0.7943	0.7608	–	–	0.8496	0.8399
MIGCN	0.7331	0.7212	0.7659	0.7244	0.8950	0.7197	–	–
IDGNN	0.7594	0.7437	0.7807	0.7417	–	–	0.7954	0.7811
SenticGAT	–	–	0.7633	0.7260	**0.9042**	0.7389	–	–
DGEDT	0.7480	0.7340	0.7680	0.7230	–	–	–	–
DGCN	0.7487	0.7227	0.7570	0.7257	0.8993	0.7488	–	–
TCKGCN	0.7592	0.7426	0.7850	0.7421	**0.9037**	0.7179	0.8920	0.8755
BERT	0.7592	0.7518	0.7759	0.7328	**0.9010**	0.7416	–	–
RGAT + BERT	0.7615	0.7488	0.7821	0.7407	**0.8971**	0.7662	–	–
HDGCN + BERT	0.7734	0.7612	0.8180	0.7888	–	–	–	–
Sentic-GCN + BERT	0.7618	0.7523	0.8147	0.7809	**0.9025**	0.7457	–	–
DualGCN + BERT	0.7740	0.7602	–	–	–	–	–	–
SGGCN + BERT	–	–	**0.8280**	**0.8020**	**0.9052**	0.7453	–	–
TFMPHGNN (ours)	0.7188	0.7064	0.7005	0.7374	0.6366	0.6294	**0.9387**	**0.9296**

Note: Bold values denote the best performance, and "–" indicates missing results in the original papers.



(a) Accuracy (b) AUC

Fig. 8. Epoch-wise training and testing curves for TFMPHGNN.



(a) VAE dimensionality (b) GCN layer

Fig. 9. Results analysis for additional parameters in TFMPHGNN.

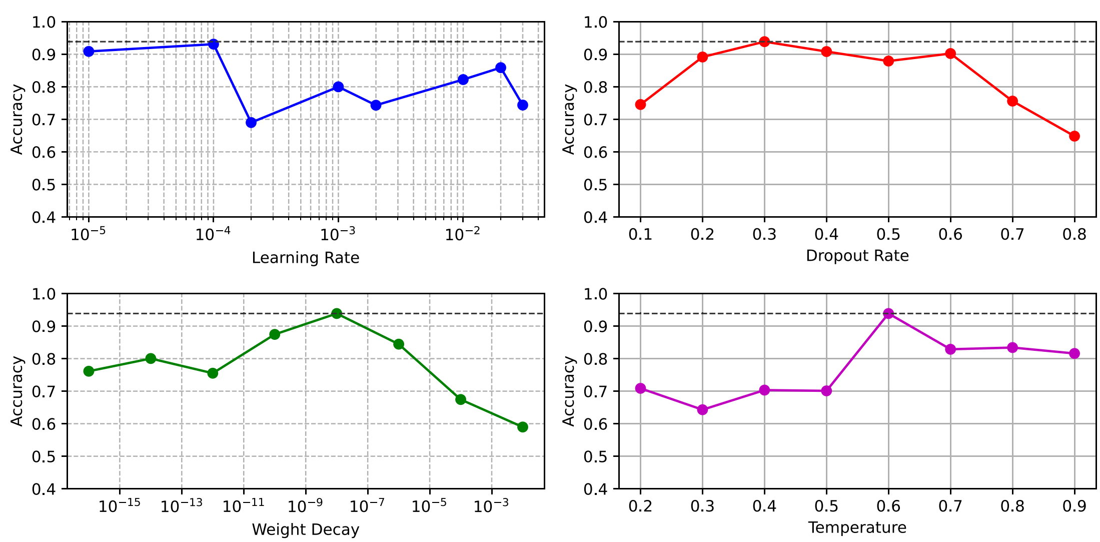


Fig. 10. Effects of additional hyperparameters.

adaptive, hierarchical feature fusion. Unlike traditional pooling, it preserves spatial and relational hierarchies among features, allowing the model to emphasize significant sentiment-emotion-context associations while mitigating overfitting. Meta-path diversity, on the other hand, enriches the model's representation by integrating multiple relational perspectives across heterogeneous graph structures. This ensures contextual completeness and improves discrimination between nuanced sentiment classes. Together, capsule routing and meta-path diversity synergistically provide robust, semantically rich embeddings that drive TFMPHGNN's superior accuracy, generalization, and interoperability. To further validate TFMPHGNN, we present epoch-wise training and testing curves that show consistent convergence and no overfitting (see Fig. 8). The testing accuracy, as shown, stabilizes around 0.93 after 35 epochs, confirming balanced performance. Additional hyperparameter analysis (see Fig. 9) demonstrates that feature dimensionality and layer depth significantly influence performance, with optimal results at a VAE dimension of 64 and two GCN layers. To further clarify the contribution of each graph perspective, a per-channel experiment is conducted. The

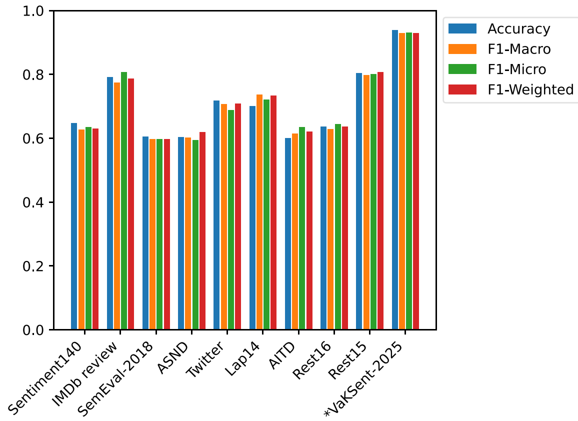


Fig. 11. Performance of TFMPHGNN with VaKSent-2025 and known sentiment datasets.

per-channel evaluation demonstrates that the topological-only configuration achieves an accuracy of 0.8421, reflecting its ability to capture structural co-occurrence patterns. In contrast, the semantic-only configuration attains 0.8635, indicating the added benefit of similarity-based relational information. The collaborative fusion channel yields a substantially higher accuracy of 0.9387, representing a 7.5% improvement over the topological channel and a 7.9% improvement over the semantic channel, thereby confirming the synergistic advantage of integrating multiple relational perspectives within the TFMPHGNN framework.

4.6.1. *Effects of additional hyperparameters*

The analysis investigates the influence of additional hyperparameters, specifically the dimensionalities of the VAE (set to 32, 64, 128, and 256) and the number of GCN layers (ranging from 1 to 6), on the performance of TFMPHGNN to enhance feature representations (see Fig. 9). The VAE's feature dimension and the GCN layer count significantly impact the model's performance. Consequently, extensive experiments were conducted to evaluate the robustness and stability of TFMPHGNN under various configurations. As illustrated in Fig. 9(a), the optimal VAE dimensionality was found to be 64, achieving an accuracy of 0.9387, an F1-macro score of 0.9296, an F1-weighted score of 0.9303, and an AUC score of 0.9400, outperforming the alternative settings of 32, 128, and 256. This finding indicates that a 64-dimensional latent space effectively balances representational capacity with computational efficiency, capturing sufficient information from the input features while mitigating the risks of overfitting and underfitting. Additionally, the impact of varying the number of GCN layers was examined, as shown in Fig. 9(b). The results reveal that TFMPHGNN's performance improves with increasing layer count, peaking at two layers; beyond this point, the model's performance declines, likely because additional layers generate increasingly similar node representations, which adversely affects node classification tasks. Finally, a range of hyperparameters—including learning rates, dropout rates, weight decay coefficients, and temperature settings—were systematically evaluated to identify the hyperparameters listed in Table 3. As shown in Fig. 10, TFMPHGNN's performance is sensitive to these hyperparameter choices, which directly affect the model's trade-offs among stability, regularization, and generalization. This careful tuning is essential for achieving state-of-the-art performance in sentiment analysis tasks within heterogeneous networks. The identified optimal settings provide a strong configuration that maximizes accuracy and enhances the model's discriminative power, thereby highlighting the crucial role of hyperparameter optimization in the overall effectiveness of TFMPHGNN.

4.6.2. *Case study*

A case study was conducted using multiple sentences from diverse sentiment texts to examine whether the TFMPHGNN model, which is based on multi-perspective meta-paths, could effectively capture diverse feature relationships for sentiment analysis tasks. The sentences, arranged from A to J, were analyzed using IDGNN, TCKGCN, DualGCN, HM-GNN, and TFMPHGNN. Prediction results are presented in Table 5, along with their corresponding truth labels for these sentences. While HM-GNN fails to correctly predict the exact polarities for all sentences, IDGNN and DualGCN predict only part of the sentences, demonstrating their inability to comprehend and fully capture semantics; they primarily analyze only partial semantic elements. In contrast, TFMPHGNN accurately predicts all polarities by incorporating multi-perspective meta-paths. This implies that, for the word "pizza," TFMPHGNN quickly determines that "service" is significant. In example D, the sentence conveys a positive sentiment, expressing relief and amusement about not having exams for the next two days. The positive phrasing includes "The good thing is" and the lighthearted "Haha!" However, TFMPHGNN fails to fully capture semantic features, possibly due to sentence ambiguity. The case studies show that TFMPHGNN excels at capturing mixed sentiments and detailed relationships in text, outperforming baseline models in all cases presented in the table. Its ability to leverage meta-path-based graph encoders and capsule networks allows it to effectively model complex interactions in heterogeneous networks, making it well-suited for sentiment analysis tasks. However, occasional errors (e.g., Case I) suggest there is room for improvement in addressing ambiguous contexts. These results demonstrate that TFMPHGNN consistently achieves state-of-the-art performance across diverse cases, establishing a new benchmark for sentiment analysis tasks that require a detailed understanding of sentiment, expressions, and context. Its strengths in managing mixed and neutral sentiments underscore its robustness and effectiveness in real-world applications.

Furthermore, nine benchmark sentiment datasets were analyzed using the proposed TFMPHGNN to evaluate its generalizability, as shown in Fig. 11. The higher F1-micro scores observed indicate class imbalance across all datasets. Meanwhile, VaKSent-2025 demonstrates itself as a clear top performer, achieving nearly perfect scores across all metrics. To this end, TFMPHGNN offers tangible benefits for SMEs and real-world applications. It can power sentiment-aware recommender systems, feedback classifiers for local businesses, and multilingual support chatbots in low-resource environments due to its modular design. Its performance under class imbalance also makes it suitable for fraud detection and mining healthcare feedback. The study faces computational demands stemming from meta-path-based capsule encoding and the limitations of multi-channel GCNs in scaling to massive, heterogeneous graphs. Additionally, lexicon dependency presents a challenge, as the quality and coverage of sentiment-emotion lexicons directly affect node feature representations and influence predictive performance. Reliance on specific dataset structures and general-purpose lexicons further limits TFMPHGNN's adaptability to new domains. Incorporating optimized sparse matrix operations, transfer learning, and domain-adaptation strategies, however, will help alleviate these challenges. A detailed comparison of the results of recent models is presented in Table 6. The table shows TFMPHGNN's state-of-the-art performance on the heterogeneous VaKSent-2025 dataset, outperforming other baselines, including the recent TCKGCN, by 4.67%. However, the model underperforms on homogeneous datasets—Twitter, Lap14, and Rest16—where simpler architectures such as SGGCN + BERT and HDGCN + BERT perform better. This disparity reflects a fundamental trade-off: TFMPHGNN's multiple meta-path types and capsule fusion are optimized for multi-type node interactions, but they introduce unnecessary overhead on homogeneous sentence graphs where meta-paths degenerate into self-loops. These results validate TFMPHGNN's robustness for multi-source sentiment analysis. Future work will investigate adaptive mechanisms that dynamically configure the model architecture in scenarios where heterogeneous structural information is unavailable.

5. Conclusion

This study introduced TFMPHGNN, a unified heterogeneous graph framework designed to advance sentiment analysis beyond conventional single-view models. TFMPHGNN integrates a meta-path–guided encoder enhanced with CapsNet and VAE alongside a multi-channel GCN; the proposed architecture effectively captures complex interactions among sentiment expressions, contextual signals, and emotional cues. This dual-network design enables richer representation learning and more discriminative feature integration across multiple sentiment dimensions. Comprehensive experiments demonstrate that TFMPHGNN achieves consistent and statistically significant improvements over state-of-the-art graph-based baselines, with gains of 4.67% in accuracy, 2.7% in F1-micro, and 4.2% in F1-weighted scores. Ablation analyses confirm that capsule dynamic routing and meta-path diversity play critical roles in enhancing representation quality, contextual awareness, and generalization. Beyond its technical contributions, TFMPHGNN offers practical value through its modular and adaptable design, supporting applications such as customer feedback analysis, sentiment-aware recommendation, and imbalanced classification scenarios. Future work will focus on improving scalability and interpretability, incorporating online learning for dynamic sentiment adaptation, integrating external knowledge graphs, and extending the framework to related tasks such as hate speech detection and emotional trigger identification. This work highlights the promise of heterogeneous graph-based deep learning as a powerful and flexible paradigm for next-generation sentiment analysis.

CRediT authorship contribution statement

Victor Kwaku Agbesi: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization; **Wenyu Chen:** Supervision, Resources, Project administration; **Chukwuebuka J. Ejiyi:** Writing – review & editing, Validation, Software, Investigation; **Gertrude Selase Gosu:** Visualization, Software, Data curation; **Chiagoziem C. Ukwuoma:** Writing – review & editing, Visualization, Resources, Formal analysis; **Olusola Bamisile:** Writing – review & editing, Supervision, Project administration.

Data availability statement

Has data associated with your study been deposited into a publicly available repository? No, the datasets used are included in the article, supplementary material, or referenced in the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was partially supported by the Key Project of National Natural Science Foundation of China Enterprise Joint Fund (Grant No. U22B2061) and the Sichuan Provincial Natural Science Foundation Project (Grant No. 61976043).

Appendix A. Supplementary data

The datasets used in this study are publicly available, freely accessible, and can be downloaded from the following: Sentiment-140: https://www.kaggle.com/datasets/kazanova/sentiment140, IMDb review is https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews, SemEval-2018: https://www.kaggle.com/datasets/context/semeval-2018-task-ec, NRC-Emotion: https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm, Subjectivity Lexicon: https://www.cs.cornell.edu/people/pabo/movie-review-data/, AITD: https://github.com/shammur/Arabic_news_text_classification_datasets/tree/main/Arabic_Influencer_Twitter_Dataset_AITD, ASND: https://github.com/shammur/Arabic_news_text_classification_datasets/tree/main/Arabic_Social_Media_News_Dataset_ASND, Lap14 https://www.kaggle.com/datasets/charitarth/semeval-2014-task-4-aspectbasedsentimentanalysis, Rest15 https://www.kaggle.com/datasets/jahanbinkia/semeval-2015-absa-restaurants, Rest16 https://www.kaggle.com/datasets/fouadaurag20/semeval-2016-absa-task5. This study is implemented in Python using the PyTorch framework. However, the codes associated with this study will be made available after publication (https://github.com/VictorAgbesi/TFMPHGNN).

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